



On some upper bounds on the intersection number of a finite soluble group

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Abstract. For a finite group G , the *intersection number* $\alpha(G)$ of G is the minimal number of maximal subgroups of G whose intersection coincides with $\Phi(G)$, the Frattini subgroup of G . In this paper, new upper bounds on $\alpha(G)$ are established when G is soluble.

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1. Introduction. All groups considered in this paper are finite.

If G is a group, following [5] (see also [10]), we denote by $\alpha(G)$ the minimal number of maximal subgroups of G whose intersection coincides with $\Phi(G)$; $\alpha(G)$ is called the *intersection number* of G (see [1]).

Let $\lambda(G)$ be the number of chief factors in a given chief series of a group G . The following upper bound was shown in [5].

Theorem 1 ([5, Theorem 3]). *If G is a soluble group, then $\alpha(G) \leq \lambda(G)$.*

Let $F(G)$ denote the Fitting subgroup of G , that is, the largest nilpotent normal subgroup of G . The following upper bound of $\alpha(G)$ was obtained in [8]. Since this paper is written in Russian, we consider it convenient to include the proof of that result here for the sake of completeness.

Theorem 2 ([8, Theorem 1.1]). *Let G be a soluble group. Assume that n is the length of a given G -chief series of the group $F(G)/\Phi(G)$, and k is the number of central G -chief factors of such a series. Then $\alpha(G) \leq 4n - 3k$.*

The following example shows that the upper bound in Theorem 2 improves the one in Theorem 1 in some cases.

Example 1. Let $n \geq 2$ and let $p_1, p_2, \dots, p_n$ be primes such that $p_1 < p_2 < \dots < p_n$. Assume that G_1 is a cyclic group of order p_1 . For $i > 1$, define G_i inductively as the semidirect product $G_i = [V_{p_i}]G_{i-1}$, where V_{p_i} is a faithful and irreducible G_{i-1} -module over the field F_{p_i} of p_i elements (such an irreducible module exists by [6, Corollary B.11.8]). Then

$$1 = G_0 \subset G_1 \subset \dots \subset G_n = G$$

is a chief series of G , $\lambda(G) = n$, $F(G) = V_{p_{n-1}}$ is a minimal normal subgroup of G and $\Phi(G) = 1$. If $n \geq 5$, then $\alpha(G) \leq 4 < n$, by Theorem 2.

Combining Theorems 1 and 2, we have:

Corollary 1. *Let G be a soluble group. Assume that n is the length of a G -chief series of the group $F(G)/\Phi(G)$, and k is the number of central G -chief factors of this series. Then*

$$\alpha(G) \leq \min \{ \lambda(G), 4n - 3k \}.$$

Let H/K be a chief factor of a group G . A subgroup A is a *complement* of H/K in G if $G = AH$ and $A \cap H = K$. We say that H/K is a *Frattini chief factor* of G if H/K is contained in the Frattini subgroup of G/K . It is clear that if H/K is abelian, then H/K is a non-Frattini chief factor of G if and only if H/K has a complement in G .

Let $\delta(G)$ denote the number of non-Frattini chief factors in a given chief series of a group G . It is well-known that $\delta(G)$ is independent of the choice of the chief series of G (see [6, Theorem A.9.13]).

It is noted in [5] that the difference $\alpha(G) - \delta(G)$ can be arbitrarily large. However, as a consequence of [5, Theorem 4], we have that

$$\alpha(G) \leq 4\delta(G)$$

for soluble groups G .

This bound can be improved with the help of prefrattini subgroups.

Let Σ be a Hall system of a soluble group G (see [6, Chapter I, Section 1.4]). By [6, Theorem I.6.6], Σ reduces in exactly one member of every conjugacy class of maximal subgroups of G . The intersection W of all complements of the non-Frattini chief factors of G in a given chief series of G into which Σ reduces is called the *prefrattini subgroup* of G associated with Σ (see [4, Section 4.3] for details). According to [9, Theorem 1.1] (see also [3, Corollary 5]), there exist elements $x, y \in G$ such that $W \cap W^x \cap W^y = \Phi(G)$. Consequently, we have

$$\alpha(G) \leq 3\delta(G)$$

for soluble groups G .

Our main goal here is to present a new upper bound on the intersection number of a soluble group in terms of the number of non-Frattini chief factors of the largest supersoluble quotient of the group.

Let $\Phi_{pri}(G)$ be the intersection of all maximal subgroups M of a group G such that $|G : M|$ is a prime, with the usual convention that $\Phi_{pri}(G) = G$ if every maximal subgroup of G has composite index. Furthermore, we denote by $\Phi_{com}(G)$ the intersection of all maximal subgroups of a group G such that

$|G : M|$ is composite ($\Phi_{com}(G) = G$ if every maximal subgroup of G has prime index). This subgroup coincides with the subgroup $L_{\mathfrak{U}}(G)$ studied in [2], where $\mathfrak{U}$ is the saturated formation of all supersoluble groups. According to [2, Corollary 3.4], $\Phi_{com}(G)$ is a supersoluble subgroup of G .

Note that every group G has a smallest normal subgroup N with supersoluble quotient G/N . This subgroup is denoted by $G^{\mathfrak{U}}$ and is called *the supersoluble residual* of the group G .

We prove the following:

Theorem 3. *Let G be a soluble group. Assume that m is the length of the G -chief series of $F(G/\Phi_{com}(G)) \cap (G/\Phi_{com}(G))^{\mathfrak{U}}$. Then*

$$\alpha(G) \leq 3\delta(G/G^{\mathfrak{U}}) + 4m.$$

Denote $\beta(G) = 3\delta(G/G^{\mathfrak{U}}) + 4m$, where m is as in Theorem 3. If G is supersoluble, then $G^{\mathfrak{U}} = 1$ and so $\beta(G) = 3\delta(G)$. However, $\beta(G)$ can be very much smaller: if G is a group as in Example 1 with p_1 not dividing $p_2 - 1$, then $\Phi_{com}(G) = 1$, $G^{\mathfrak{U}} = V_{p_n} \cdots V_{p_2}$ and $m = 1$. Then $\beta(G) = 7$ and $3\delta(G) = 3n$. Hence $\beta(G) < 3n$ if $n \geq 3$.

The following example shows that in some cases $\beta(G)$ is smaller than $\min\{\lambda(G), 4n - 3k\}$, where $4n - 3k$ is the bound determined in Theorem 2.

Example 2. Let p, q , and r be three primes such that p divides $q - 1$ and p does not divide $r - 1$. Let A be the cyclic group of order p^c for some $c \geq 1$. Let V_q be an irreducible A -module over the field of q -elements such that $B = C_A(V_q)$ is the maximal subgroup of A . Then V_q is one-dimensional. Let V_r be an irreducible A -module over the field of r elements with $B = C_A(V_r)$. Then V_r is not one-dimensional. Let V be a direct sum of a copies of V_q and let W be a direct sum of b copies of V_r . Consider $G = [V \oplus W]A$, the corresponding semidirect product. Then $G^{\mathfrak{U}} = W$, $F(G) = [V \oplus W]B$, and $G/G^{\mathfrak{U}} \cong [V]A$. Furthermore, G has a core-free maximal subgroup of composite index. Therefore $\Phi_{com}(G) = 1$. Hence $\beta(G) = 3a + 3 + 4b$, $\lambda(G) = a + b + c$, and the upper bound determined in Theorem 2 is equal to $4a + 4b$. If we take $a = 4$, $b = 1$, and $c = 15$, it follows that $\beta(G) = 19$, $\lambda(G) = 20$, and $4a + 4b = 20$.

2. Preliminary results. We begin with an elementary lemma. Its proof is straightforward.

Lemma 1. *Let N be a normal subgroup of a group G .*

- (1) *If $N \subseteq \Phi_{pri}(G)$, then $\Phi_{pri}(G/N) = \Phi_{pri}(G)/N$.*
- (2) *If $N \subseteq \Phi_{com}(G)$, then $\Phi_{com}(G/N) = \Phi_{com}(G)/N$.*
- (3) *$\Phi(G/\Phi_{com}(G)) = 1$.*
- (4) *$\Phi(G) = \Phi_{pri}(G) \cap \Phi_{com}(G)$.*

Lemma 2. *If G is a soluble group, then $G^{\mathfrak{U}} \subseteq \Phi_{pri}(G)$ and $\Phi_{pri}(G)/G^{\mathfrak{U}} = \Phi(G/G^{\mathfrak{U}})$.*

Proof. Let M be a maximal subgroup of G such that $|G : M| = p$, where p is a prime. Then $G/\text{Core}_G(M)$ is primitive, and since G is soluble, M is a complement of a G -chief factor $N/\text{Core}_G(M)$ in G by [6, Theorem A.15.2]. Since

$|G : M| = |N/\text{Core}_G(M)|$, it follows that $N/\text{Core}_G(M)$ has order p . By [6, Theorem A.21.1], $G/C_G(N/\text{Core}_G(M))$ is cyclic of order dividing $p - 1$. Since $C_{G/\text{Core}_G(M)}(N/\text{Core}_G(M)) = N/\text{Core}_G(M)$, we conclude that $G/\text{Core}_G(M)$ is supersoluble. Hence $G^\mathfrak{U} \subseteq \text{Core}_G(M) \subseteq M$ and $G^\mathfrak{U} \subseteq \Phi_{\text{pri}}(G)$.

Since $G/G^\mathfrak{U}$ is supersoluble, all maximal subgroups of $G/G^\mathfrak{U}$ have prime index by [6, Theorem VII.2.2]. Hence $\Phi_{\text{pri}}(G)/G^\mathfrak{U} = \Phi_{\text{pri}}(G/G^\mathfrak{U}) = \Phi(G/G^\mathfrak{U})$. $\square$

Lemma 3. *Let G be a group. Then $\alpha(G) \leq \alpha(G/\Phi_{\text{pri}}(G)) + \alpha(G/\Phi_{\text{com}}(G))$.*

Proof. Suppose that $\alpha(G/\Phi_{\text{pri}}(G)) = k$. Since $\Phi_{\text{pri}}(G/\Phi_{\text{pri}}(G)) = 1$ by Lemma 1, there exist maximal subgroups $M_1, M_2, \dots, M_k$ of G such that, for every $i = 1, 2, \dots, k$, $|G : M_i|$ is prime and

$$\Phi_{\text{pri}}(G) = M_1 \cap M_2 \cap \dots \cap M_k.$$

Assume that $\alpha(G/\Phi_{\text{com}}(G)) = n$. Since $\Phi_{\text{com}}(G/\Phi_{\text{com}}(G)) = 1$ by Lemma 1, there exist maximal subgroups $H_1, H_2, \dots, H_n$ of G such that, for every $j = 1, 2, \dots, n$, $|G : H_j|$ is a composite number and

$$\Phi_{\text{com}}(G) = H_1 \cap H_2 \cap \dots \cap H_n.$$

Then, by Lemma 1, we have

$$\Phi(G) = (M_1 \cap M_2 \cap \dots \cap M_k) \cap (H_1 \cap H_2 \cap \dots \cap H_n).$$

Hence $\alpha(G) = \alpha(G/\Phi(G)) \leq k + n$. $\square$

Lemma 4. *Let G be a soluble group. Then*

$$\alpha(G/\Phi_{\text{pri}}(G)) \leq \delta(G/G^\mathfrak{U}).$$

Proof. By Lemma 2, $G^\mathfrak{U} \subseteq \Phi_{\text{pri}}(G)$. Hence $G/\Phi_{\text{pri}}(G)$ is supersoluble. Since the derived subgroup of every supersoluble group is nilpotent, we can apply [5, Theorem 4.4] to conclude that $\alpha(G/\Phi_{\text{pri}}(G)) \leq \delta(G/G^\mathfrak{U})$. $\square$

Lemma 5 ([7, Theorem 1.4]). *Let G be a soluble group and V a finite faithful G -module. If V is completely reducible, there exist $v_1, v_2, v_3 \in V$ such that $C_G(v_1) \cap C_G(v_2) \cap C_G(v_3) = 1$.*

For abelian groups, this can be strengthened as follows.

Lemma 6 ([5, Lemma 4.2]). *Let G be an abelian group and let V be a faithful completely reducible G -module. Then there exists $x \in V$ such that $C_G(x) = 1$.*

Lemma 7 ([6, Lemma A.16.3]). *Let $G = [N]H$ be a semidirect product of a normal subgroup N with a subgroup H .*

- (a) *If $n \in N$, then $H \cap H^n = C_H(n)$.*
- (b) *$\text{Core}_G(H) = C_H(N)$.*

Lemma 8. *Let G be a soluble primitive group. Let M be a maximal subgroup of G with $\text{Core}_G(M) = 1$. Then there exist $x, y, z \in G$ such that*

$$M \cap M^x \cap M^y \cap M^z = 1.$$

If G is supersoluble, there exists $x \in G$ such that $M \cap M^x = 1$.

Proof. Since G is soluble and primitive, G has a unique minimal normal subgroup N , which is complemented by a maximal subgroup M of G . Moreover, $N = C_G(N)$ and $\text{Core}_G(M) = 1$. Hence M acts faithfully and irreducibly on N .

By Lemma 5, there exist $x, y, z \in N$ such that

$$C_M(x) \cap C_M(y) \cap C_M(z) = 1.$$

Thus, by Lemma 7, we conclude that $M \cap M^x \cap M^y \cap M^z = 1$.

If G is supersoluble, then M is abelian. By Lemma 6, there exists $x \in N$ such that $C_M(x) = 1$ and then $M \cap M^x = 1$. $\square$

Lemma 9. *Let G be a soluble group with $\Phi_{\text{com}}(G) = 1$. Suppose that n is the length of a G -chief series of the group $F(G)$, and k is the number of cyclic G -chief factors of this series. Then G contains $4n - 2k$ maximal subgroups whose intersection is $\Phi_{\text{com}}(G) = 1$.*

Proof. By Lemma 1, $\Phi(G) = 1$. Then $F(G)$ is a direct product of minimal normal subgroups of G , all complemented in G by [6, Theorem A.10.6]. Assume that

$$F(G) = (N_1 \times \cdots \times N_k) \times (N_{k+1} \times \cdots \times N_n),$$

where N_i is a minimal normal subgroup of G of prime order, for all $i = 1, \dots, k$, and N_j is a minimal normal subgroup of G of composite order, for every $j \in \{k+1, \dots, n\}$.

Consider the following G -chief series of $F(G)$:

$$1 = N_0 \subset N_1 \subset N_1 N_2 \subset \cdots \subset N_1 N_2 \cdots N_n = F(G).$$

Assume that M_i is a maximal subgroup of G which complements the G -chief factor $N_1 N_2 \cdots N_i / N_1 N_2 \cdots N_{i-1}$ for each $i = 1, \dots, n$ ($N_1 N_2 \cdots N_{i-1} = 1$ if $i = 1$). Let

$$X = \text{Core}_G(M_1) \cap \text{Core}_G(M_2) \cap \cdots \cap \text{Core}_G(M_n).$$

Assume that $X \neq 1$ and let N be a minimal normal subgroup of G contained in X . Then N is contained in $F(G)$, and there exists $i \in \{1, \dots, n\}$ such that $N_1 N_2 \cdots N_i \subseteq N(N_1 N_2 \cdots N_{i-1})$. Then $G = NM_i$. This is impossible since $N \subseteq M_i$. Hence $X = 1$, and

$$\Phi_{\text{com}}(G) = \text{Core}_G(M_1) \cap \text{Core}_G(M_2) \cap \cdots \cap \text{Core}_G(M_n) = 1.$$

By Lemma 8, we have that

$$\text{Core}_G(M_j) = M_j \cap M_j^{x_j} \cap M_j^{y_j} \cap M_j^{z_j}$$

for every $j \in \{k+1, \dots, n\}$, and $\text{Core}_G(M_i) = M_i \cap M_i^{x_i}$ for all $i = 1, \dots, k$. Hence G contains $4(n-k) + 2k = 4n - 2k$ maximal subgroups whose intersection is $\Phi_{\text{com}}(G) = 1$. $\square$

3. Proof of Theorem 2.

Proof. We argue by induction on $|G|$. Assume $\Phi(G) \neq 1$. Since $\Phi(G/\Phi(G)) = 1$ and $F(G/\Phi(G)) = F(G)/\Phi(G)$, the length of a $G/\Phi(G)$ -chief series of the group $F(G/\Phi(G))$ is equal to n , and the number of central $G/\Phi(G)$ -chief factors of this series is equal to k . By induction, there exist maximal subgroups $M_1/\Phi(G), M_2/\Phi(G), \dots, M_{4n-3k}/\Phi(G)$ of $G/\Phi(G)$ whose intersection is equal to 1. Then

$$M_1 \cap M_2 \cap \dots \cap M_{4n-3k} = \Phi(G),$$

and $\alpha(G) \leq 4n - 3k$.

Assume $\Phi(G) = 1$. By [6, Theorem A.10.6], $F(G) = N_1 \cdots N_n$ is the product of minimal normal subgroups $N_1, \dots, N_n$ of G , all complemented in G . Without loss of generality, we can assume that $N_1, N_2, \dots, N_k$ are central in G , and $N_{k+1}, N_{k+2}, \dots, N_n$ are non-central. Consider the G -chief series

$$1 = N_0 \subset N_1 \subset N_1 N_2 \subset \dots \subset N_1 N_2 \cdots N_n = F(G)$$

of $F(G)$. Assume that M_i is a maximal subgroup of G which complements the G -chief factor $N_1 N_2 \cdots N_i / N_1 N_2 \cdots N_{i-1}$ for each $i = 1, \dots, n$. Denote the subgroup

$$\text{Core}_G(M_1) \cap \text{Core}_G(M_2) \cap \dots \cap \text{Core}_G(M_n)$$

by X . As in Lemma 9, we have

$$\text{Core}_G(M_1) \cap \text{Core}_G(M_2) \cap \dots \cap \text{Core}_G(M_n) = X = 1 = \Phi(G).$$

By Lemma 8, we have that

$$\text{Core}_G(M_j) = M_j \cap M_j^{x_j} \cap M_j^{y_j} \cap M_j^{z_j}$$

for every $j \in \{k+1, \dots, n\}$. Since the subgroups $M_1, M_2, \dots, M_k$ are normal in G , $\text{Core}_G(M_i) = M_i$ for every $i = 1, 2, \dots, k$.

Hence G contains $4(n-k) + k = 4n - 3k$ maximal subgroups whose intersection is $\Phi(G)$. Therefore $\alpha(G) \leq 4n - 3k$, as requested. $\square$

4. Proof of Theorem 3.

Proof. By Lemmas 3 and 4, we have that

$$\alpha(G) \leq \alpha(G/\Phi_{\text{pri}}(G)) + \alpha(G/\Phi_{\text{com}}(G)) \leq \delta(G/G^{\mathfrak{U}}) + \alpha(G/\Phi_{\text{com}}(G)).$$

If $(G/\Phi_{\text{com}}(G))^{\mathfrak{U}} = 1$, then $G^{\mathfrak{U}}$ is contained in $\Phi(G)$ by Lemma 2 and so G is supersoluble. Hence $G^{\mathfrak{U}} = 1$, $n = 0$, $\alpha(G) \leq 3\delta(G)$, and the theorem holds. Hence we may assume that $(G/\Phi_{\text{com}}(G))^{\mathfrak{U}} \neq 1$. Since $\Phi(G/\Phi_{\text{com}}(G)) = 1$, it follows that $F(G/\Phi_{\text{com}}(G))$ is a direct product of minimal normal subgroups of $G/\Phi_{\text{com}}(G)$, all of them complemented in $G/\Phi_{\text{com}}(G)$ by [6, Theorem A.10.6]. Suppose that $N/\Phi_{\text{com}}(G)$ is a minimal normal subgroup of $G/\Phi_{\text{com}}(G)$ contained in $(G/\Phi_{\text{com}}(G))^{\mathfrak{U}}$. By Lemma 1, $\Phi_{\text{com}}(G/\Phi_{\text{com}}(G)) = 1$, and therefore there exists a maximal subgroup of G of composite index complementing $N/\Phi_{\text{com}}(G)$ in $G/\Phi_{\text{com}}(G)$. In particular, $N/\Phi_{\text{com}}(G)$ is not cyclic. Assume that k is the number of cyclic G -chief factors of a G -chief series of $F(G/\Phi_{\text{com}}(G))$. Since every G -chief factor above $G^{\mathfrak{U}}$ is cyclic, we conclude that

$m + k$ is the length of a G -chief series of $F(G/\Phi_{\text{com}}(G))$. Applying Lemma 9, we have that G has $4m + 2k$ maximal subgroups whose intersection is $\Phi_{\text{com}}(G)$. In particular, $\alpha(G/\Phi_{\text{com}}(G)) \leq 4m + 2k$. Therefore

$$\begin{aligned}\alpha(G) &= \alpha(G/\Phi(G)) \leq \alpha(G/\Phi_{\text{pri}}(G)) + \alpha(G/\Phi_{\text{com}}(G)) \\ &\leq \delta(G/G^{\mathfrak{U}}) + 4m + 2k \leq 3\delta(G/G^{\mathfrak{U}}) + 4m.\end{aligned}$$

□

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